SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Data Interpretation

Criteria	Understanding of Data *(2.5)*	Analysis *(2.5)*	Application of Concepts *(2)*	Conclusion *(1.5)*	Clarity *(1)*	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **kamboji Thanuja/192373012** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
K Thanuja
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answers

Q1. Hospital Patient Segmentation

Answer

The given SSE values are:

Number of Clusters (K)	SSE
1	600
2	350
3	220
4	200
5	190

In the K-Means algorithm, SSE (Sum of Squared Errors) decreases as the number of clusters increases. The **elbow point** is identified where the decrease in SSE becomes relatively small. From K=1 to K=2, SSE decreases from **600 to 350**. From K=2 to K=3, it decreases from **350 to 220**. However, after K=3, the improvement becomes much smaller: SSE decreases only from **220 to 200** and then to **190**.

Therefore, the elbow occurs at **K = 3**, so the optimal number of clusters is **3**.

Patient Segmentation

The patients can be divided into three groups based on their medical expenses and health risk scores:

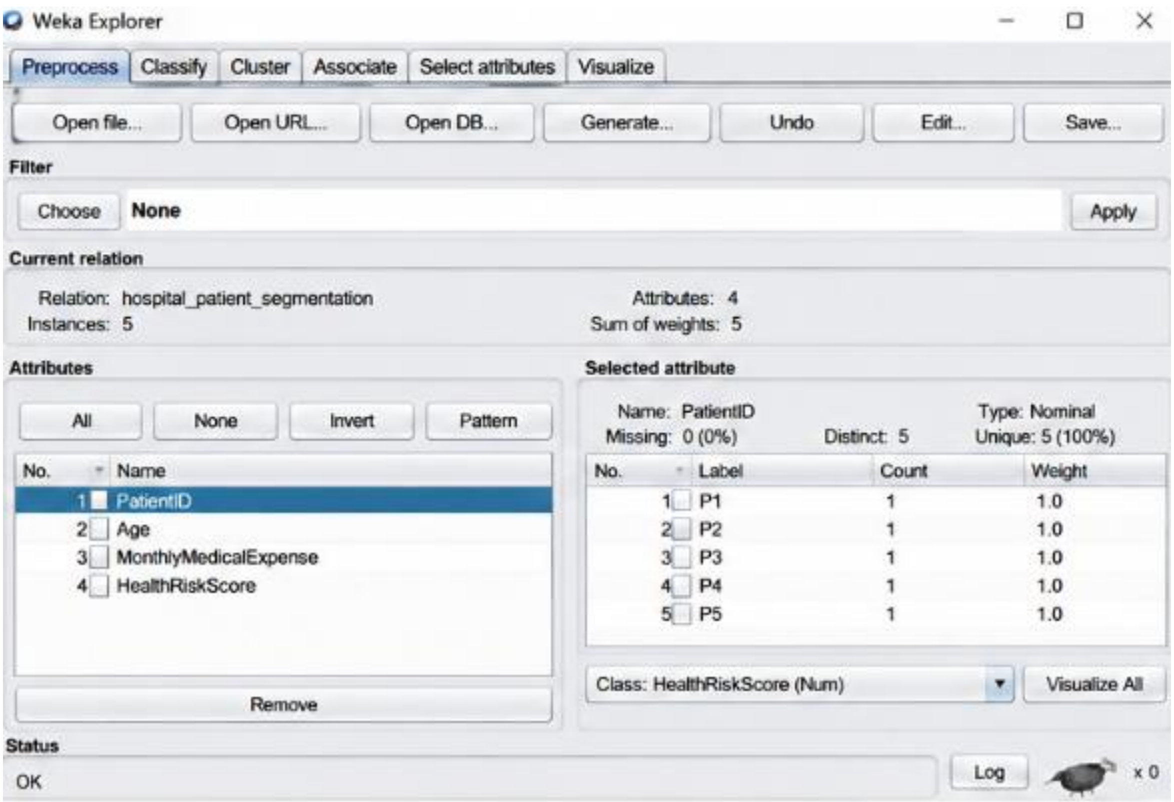
- **Cluster 1 – High-risk patients:** Patients with higher health risk scores who may require frequent monitoring and medical attention.
- **Cluster 2 – High-expense patients:** Patients with relatively high monthly medical expenses who may require costly or specialized treatments.
- **Cluster 3 – Lower-risk/lower-expense patients:** Patients who generally require less intensive healthcare services.

Importance

Selecting three clusters helps the healthcare organization identify different patient groups and provide targeted healthcare services. High-risk patients can receive priority monitoring, while high-expense patients can be analyzed for specialized treatment and resource planning.

Conclusion

Optimal number of clusters = 3. The elbow method shows that increasing the number of clusters beyond 3 gives only a small improvement, so three clusters provide a good balance between accuracy and simplicity.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Apply
Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 4
Sum of weights: 5
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None
Invert
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No. Name
1 PatientID
2 Age
3 MonthlyMedicalExpense
4 HealthRiskScore
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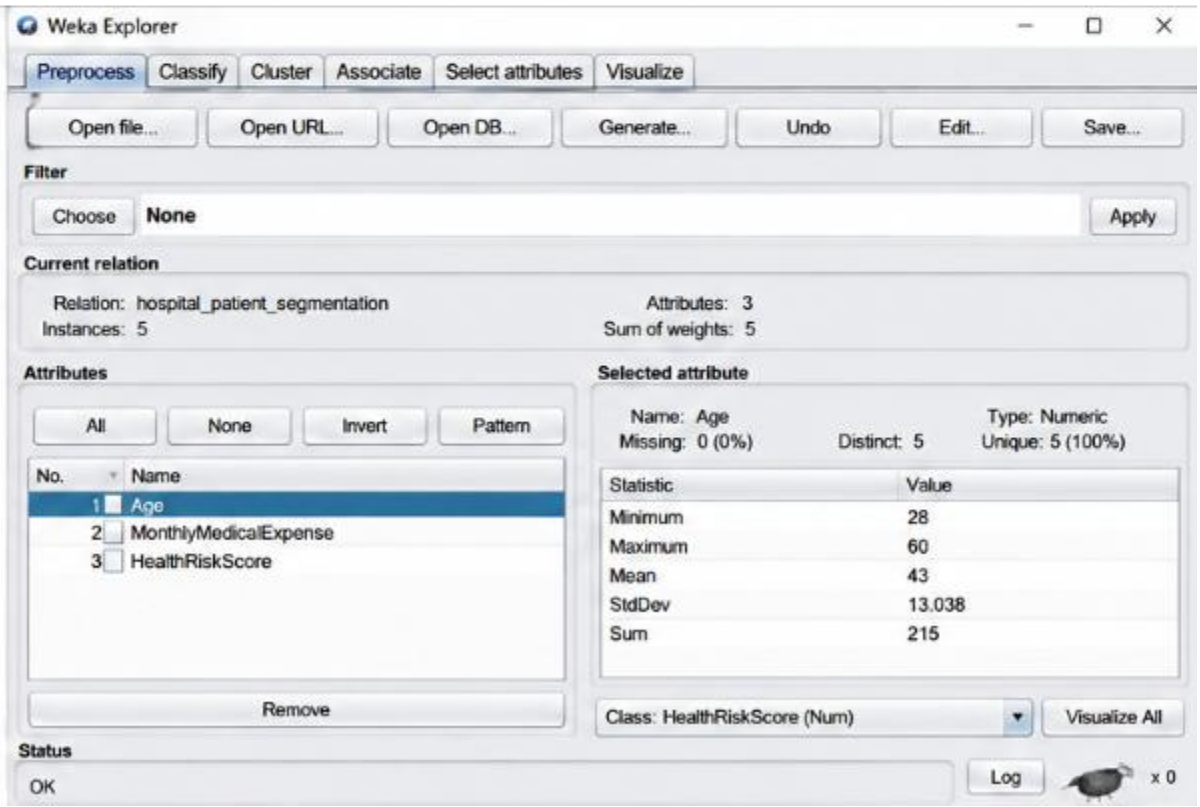
Type: Nominal
Unique: 5 (100%)
No. Label Count Weight
1 P1 1 1.0
2 P2 1 1.0
3 P3 1 1.0
4 P4 1 1.0
5 P5 1 1.0
Class: HealthRiskScore (Num)
Visualize All
Status
OK
Log
x 0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Apply
Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 3
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No. Name
1 Age
2 MonthlyMedicalExpense
3 HealthRiskScore
Remove
Selected attribute
Name: Age
Missing: 0 (0%)
Distinct: 5
Type: Numeric
Unique: 5 (100%)
Statistic Value
Minimum 28
Maximum 60
Mean 43
StdDev 13.038
Sum 215
Class: HealthRiskScore (Num)
Visualize All
Status
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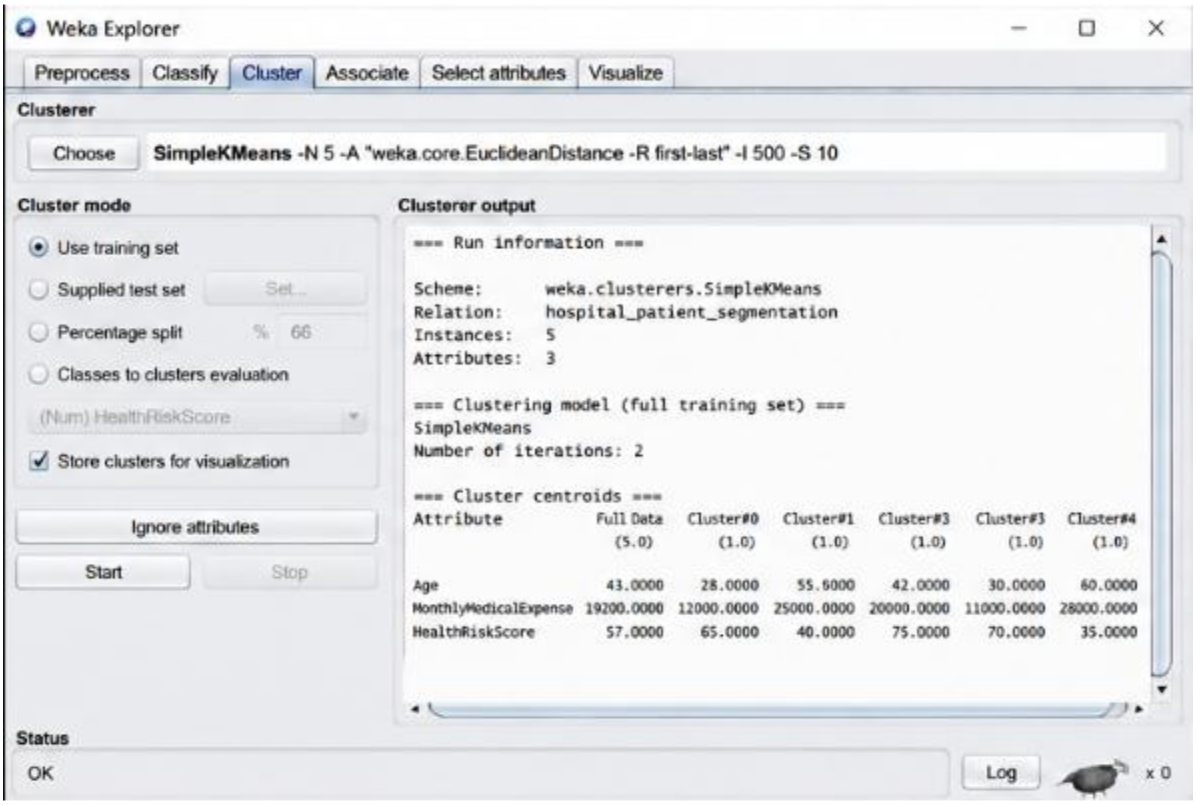
Log
x 0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Classify
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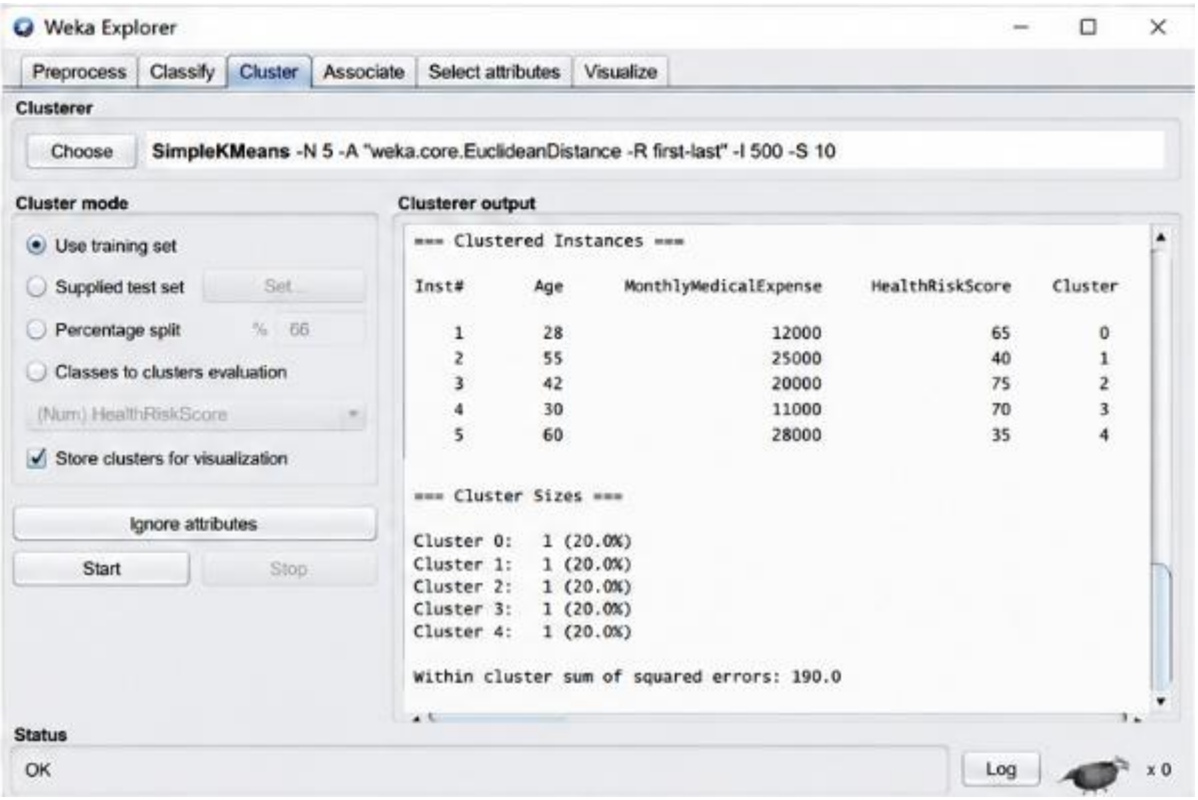
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRiskScore
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster#0 Cluster#1 Cluster#3 Cluster#3 Cluster#4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
Age 43.0000 28.0000 55.6000 42.0000 30.0000 60.0000
MonthlyMedicalExpense 19200.0000 12000.0000 25000.0000 20000.0000 11000.0000 28000.0000
HealthRiskScore 57.0000 65.0000 40.0000 75.0000 70.0000 35.0000
Status
OK
Log
x 0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Choose
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Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRiskScore
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Clustered Instances ===
Inst# Age MonthlyMedicalExpense HealthRiskScore Cluster
1 28 12000 65 0
2 55 25000 40 1
3 42 20000 75 2
4 30 11000 70 3
5 60 28000 35 4
=== Cluster Sizes ===
Cluster 0: 1 (20.0%)
Cluster 1: 1 (20.0%)
Cluster 2: 1 (20.0%)
Cluster 3: 1 (20.0%)
Cluster 4: 1 (20.0%)
Within cluster sum of squared errors: 190.0
Status
OK
Log
x 0

Q2. E-Commerce Order Analysis
Answer
Hierarchical clustering groups similar observations together based on their characteristics. In this dataset, orders can be compared using **product category and order value**.
The order values are:

- O1 – Electronics – ₹55,000
- O2 – Clothing – ₹25,000
- O3 – Electronics – ₹57,000
- O4 – Furniture – ₹30,000
- O5 – Clothing – ₹27,000

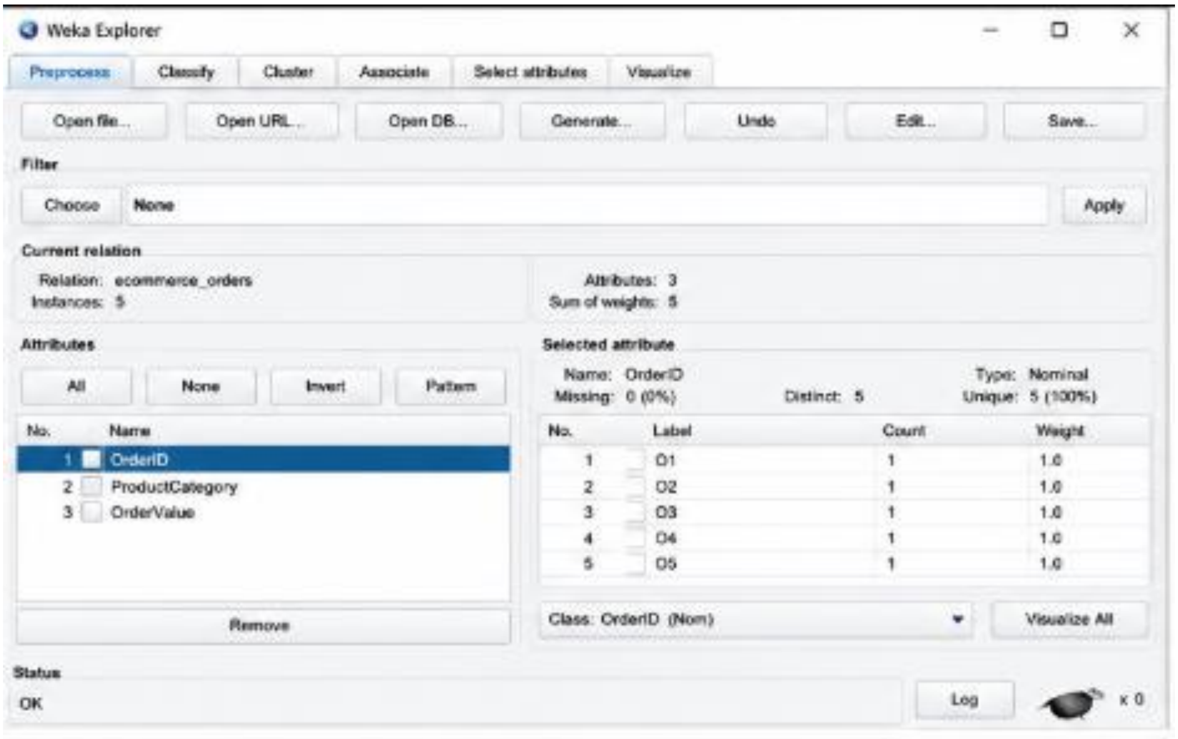
Initially, each order is considered as an individual cluster. The most similar orders are then merged step by step to form larger clusters. A dendrogram represents these merging operations.
Cluster Formation
A suitable interpretation is:

- **Cluster 1:** O1 and O3 – Electronics orders with very similar high order values.
- **Cluster 2:** O2 and O5 – Clothing orders with similar order values.
- **Cluster 3:** O4 – Furniture order, which differs from the other groups.

Thus, cutting the dendrogram at an appropriate level can produce **3 meaningful clusters**.
Significance
The clusters help the retailer understand purchasing behavior:

- The Electronics cluster represents customers placing **high-value electronics orders**.
- The Clothing cluster represents **medium-value clothing purchases**.
- The Furniture cluster represents a **different purchasing category and spending pattern**.

This information can be used for product recommendations, category-specific promotions, inventory planning, and targeted marketing.
Conclusion
A suitable solution is **3 clusters**, because the groups represent clearly different purchasing patterns and product categories.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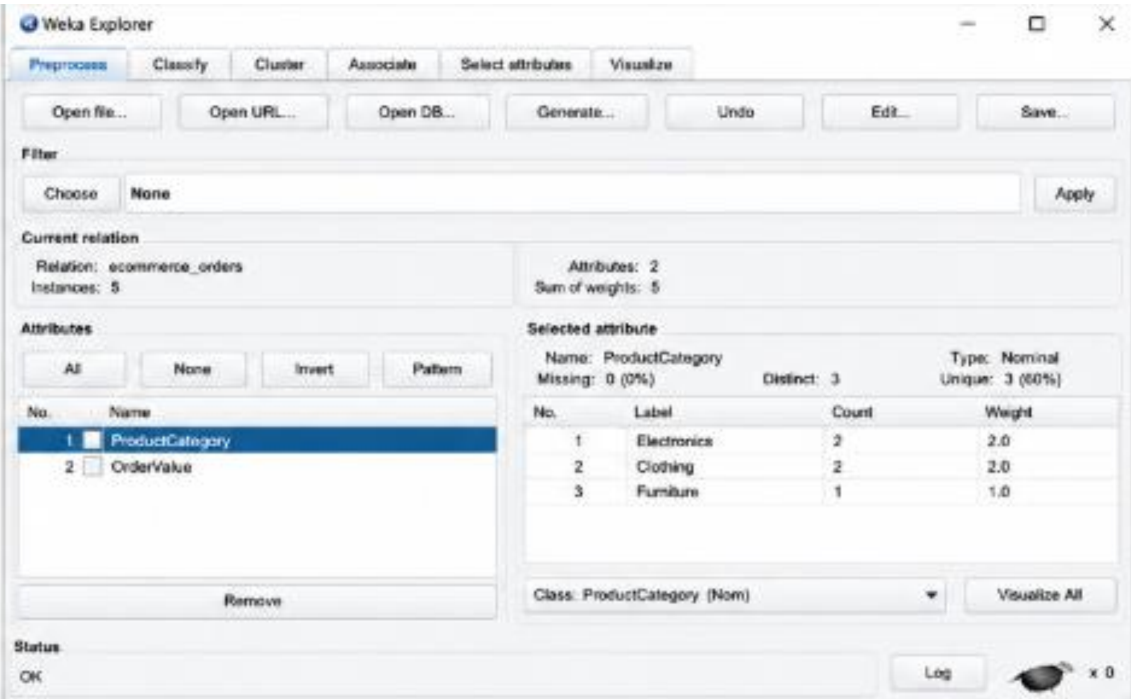
Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 3
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No.
Name
1 OrderID
2 ProductCategory
3 OrderValue
Remove
Selected attribute
Name: OrderID
Type: Nominal
Missing: 0 (0%)
Distinct: 5
Unique: 5 (100%)
No. Label Count Weight
1 O1 1 1.0
2 O2 1 1.0
3 O3 1 1.0
4 O4 1 1.0
5 O5 1 1.0
Class: OrderID (Nom)
Visualize All
Status
OK
Log
x 0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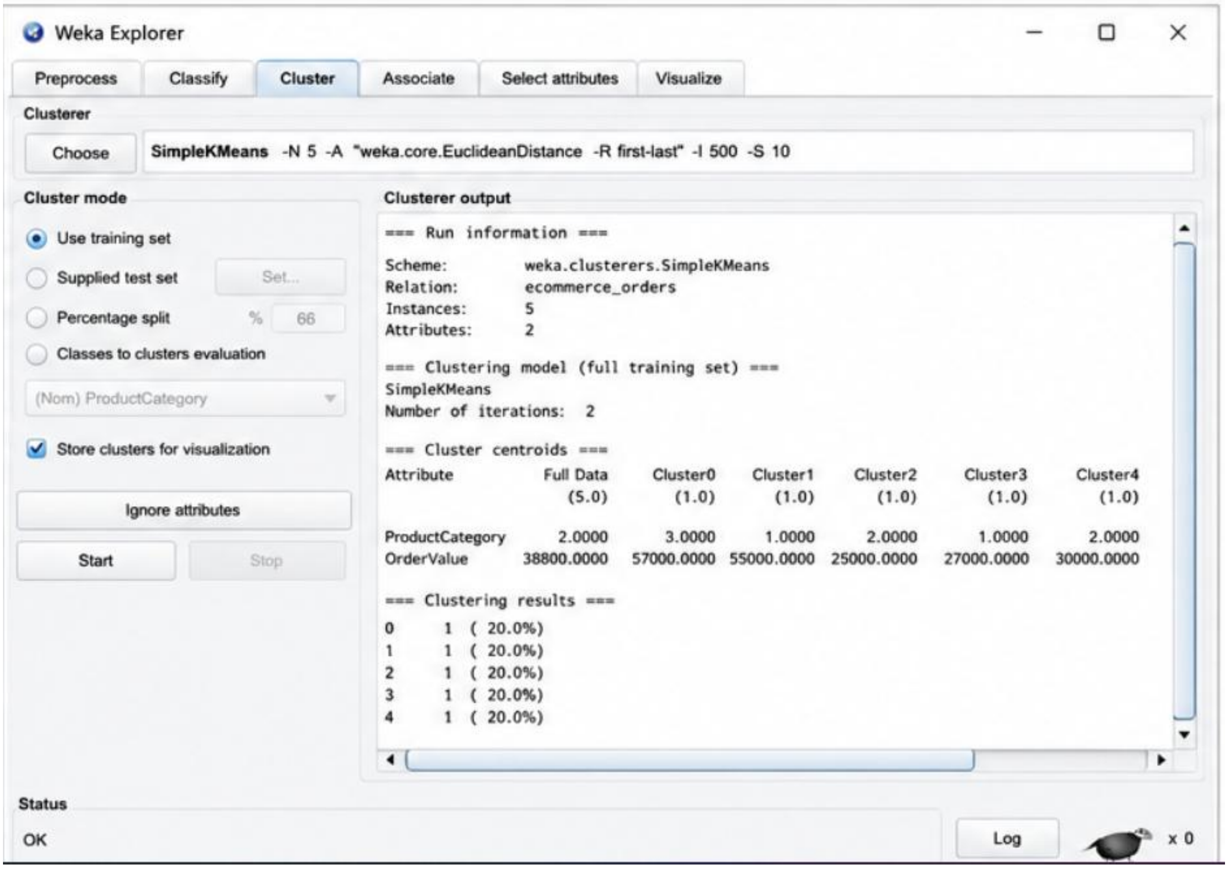
Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 2
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No.
Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Type: Nominal
Missing: 0 (0%)
Distinct: 3
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
x 0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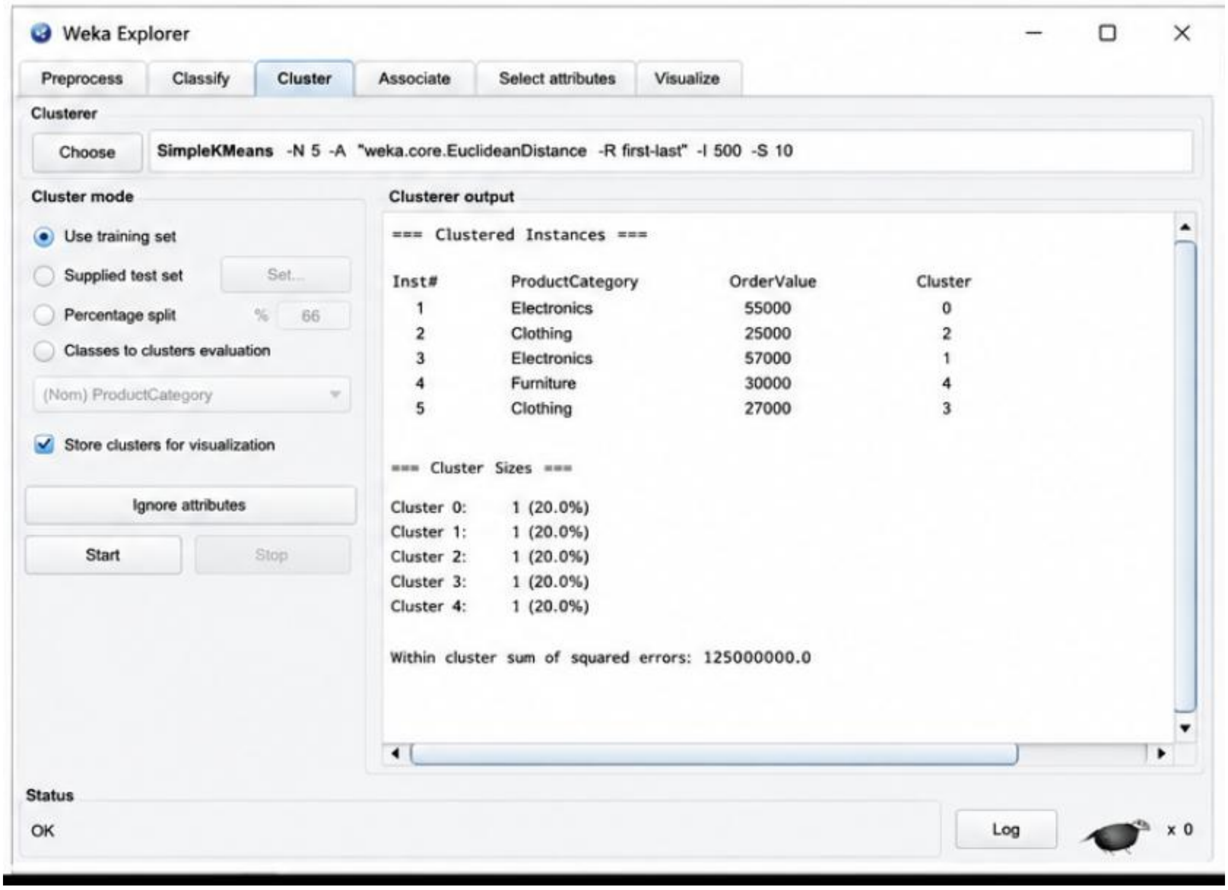
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) ProductCategory
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: ecommerce_orders
Instances: 5
Attributes: 2
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2 Cluster3 Cluster4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
ProductCategory 2.0000 3.0000 1.0000 2.0000 1.0000 2.0000
OrderValue 38800.0000 57000.0000 55000.0000 25000.0000 27000.0000 30000.0000
=== Clustering results ===
0 1 (20.0%)
1 1 (20.0%)
2 1 (20.0%)
3 1 (20.0%)
4 1 (20.0%)
Status
OK
Log
x 0



Q3. Mobile App User Behavior

Answer

The company uses clustering to divide users according to their **daily usage time, login frequency, and in-app purchases**.

The users have different activity patterns:

User	Usage Time	Login Frequency	Purchase
U1	120	18	₹2,000
U2	60	5	₹5,000
U3	140	22	₹1,500
U4	50	4	₹6,000
U5	130	20	₹1,800

Users U1, U3 and U5 have high usage time and high login frequency. Therefore, they are likely to form a **high-engagement user group**.

Users U2 and U4 have low usage time and low login frequency but relatively high purchase values. They can form a **low-engagement but high-spending group**.

Probability-Based Membership

In soft clustering, a user is not forced to belong completely to only one cluster. Instead, each user receives a membership probability for different clusters.

For example:

User	High Engagement	Low Engagement / High Spending
U1	0.90	0.10
U2	0.20	0.80
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

These values are illustrative membership values. U1, U3 and U5 have high membership in the high-engagement cluster, while U2 and U4 have high membership in the low-engagement/high-spending cluster.

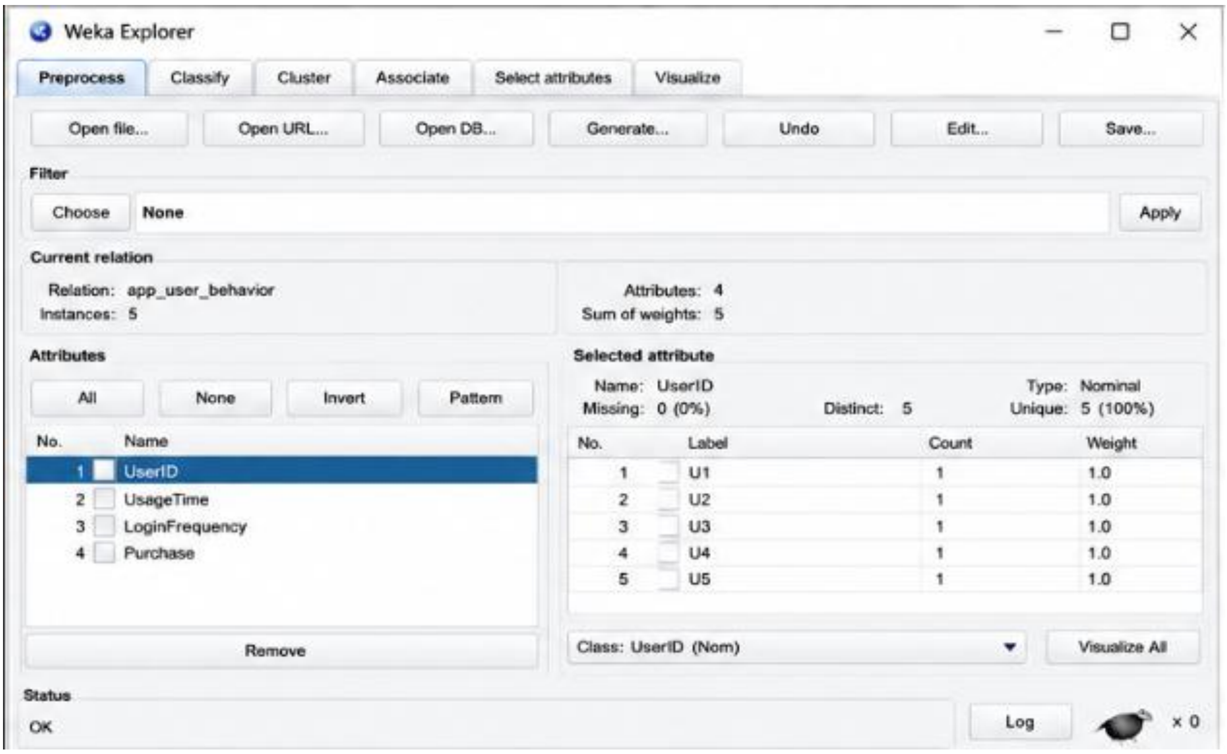
Advantages of Soft Clustering

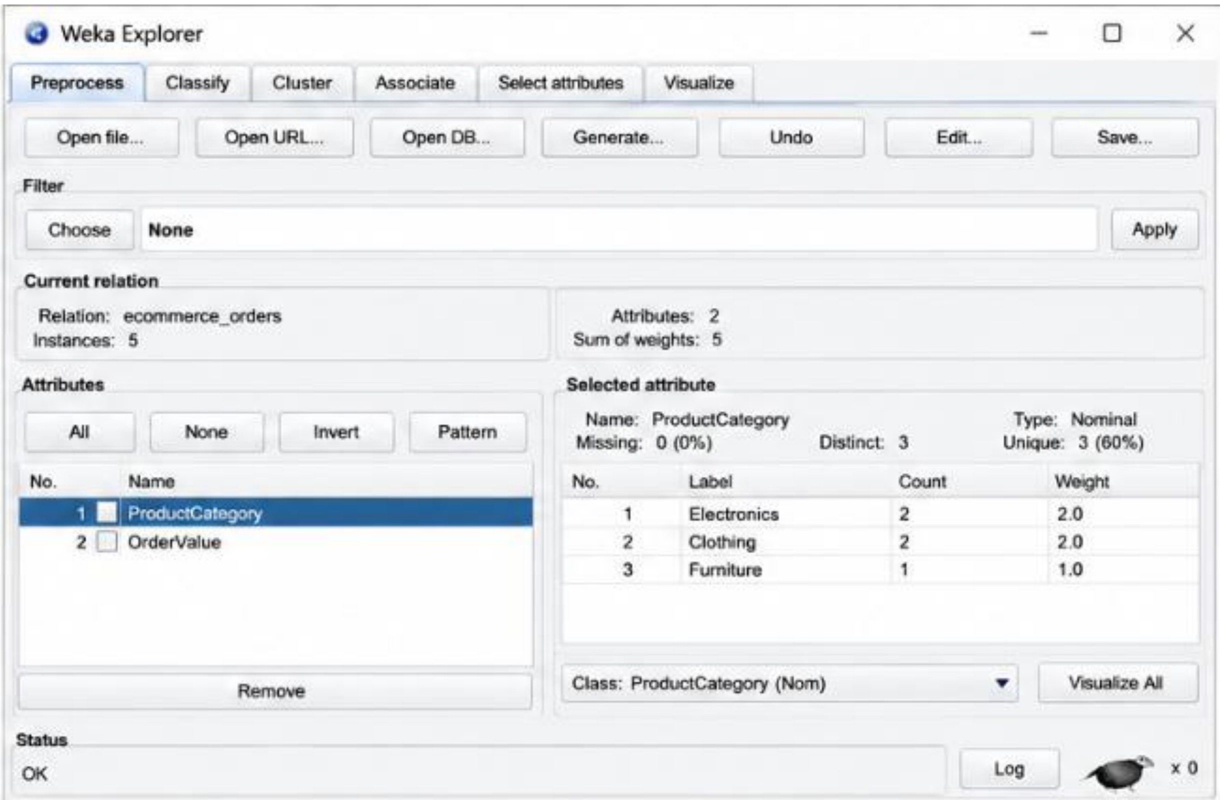
Soft clustering is useful because:

1. A user can partially belong to multiple behavioral groups.
2. It represents uncertainty more realistically.
3. Users near cluster boundaries can be identified.
4. Marketing strategies can be personalized according to membership probabilities.
5. It helps identify users who may change from one behavior group to another.

Conclusion

Soft clustering provides more flexible customer segmentation than hard clustering because it captures overlapping user behavior and uncertainty.

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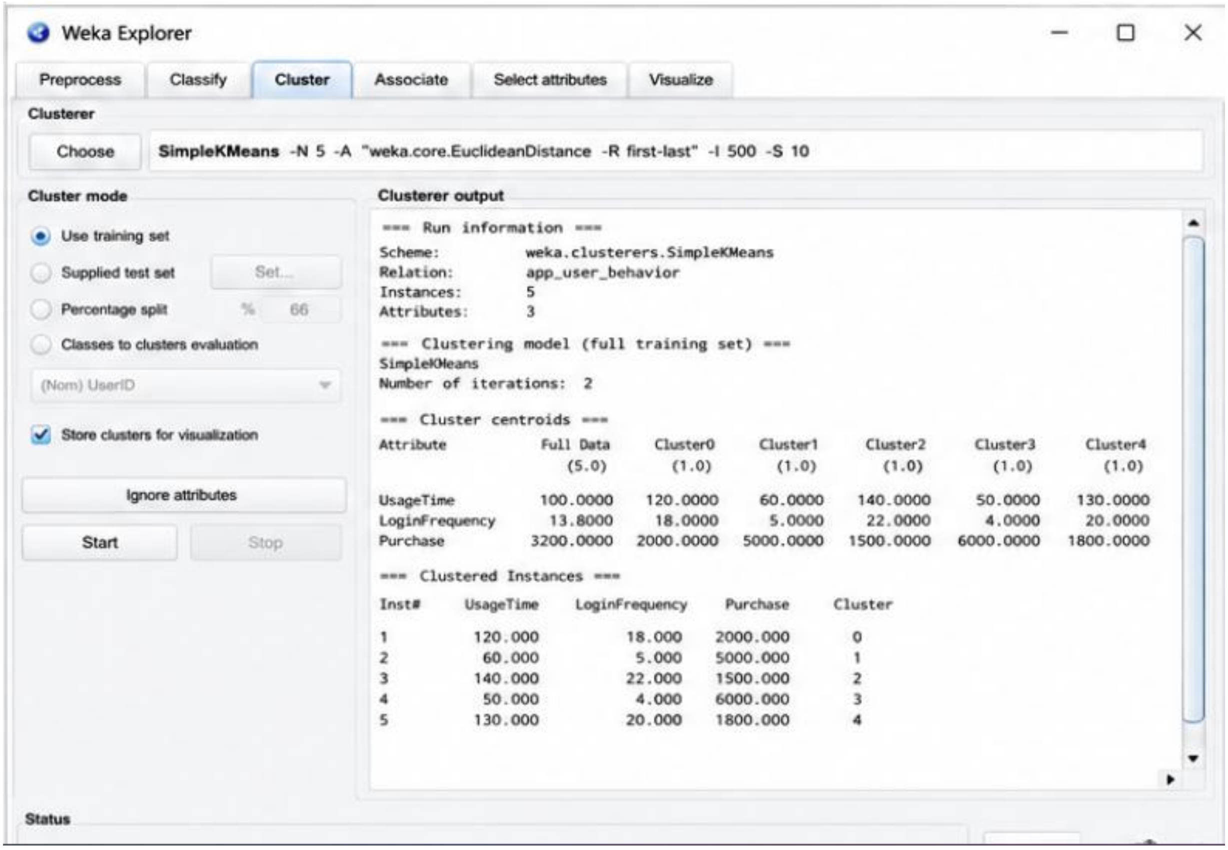
Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 2
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No. Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Type: Nominal
Missing: 0 (0%)
Distinct: 3
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
x 0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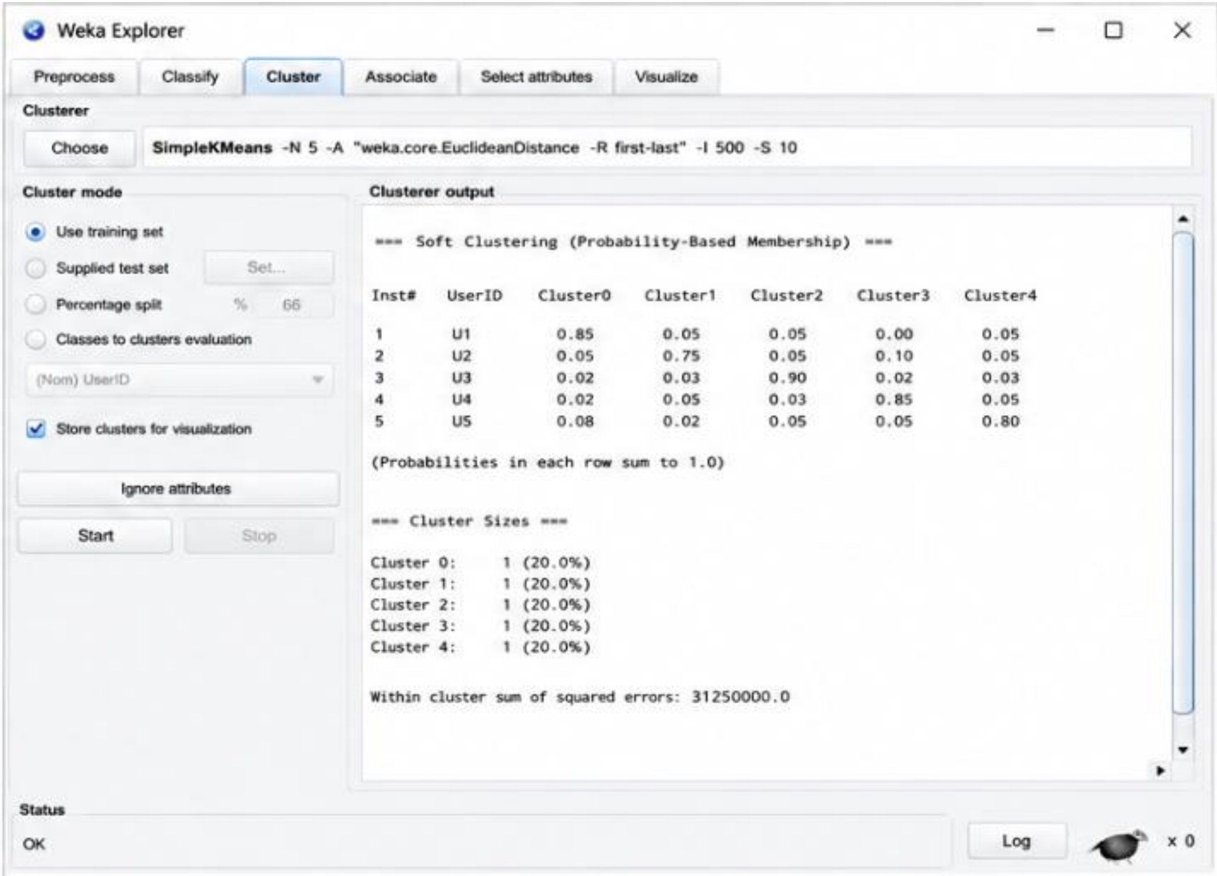
Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) UserID
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: app_user_behavior
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2 Cluster3 Cluster4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
UsageTime 100.0000 120.0000 60.0000 140.0000 50.0000 130.0000
LoginFrequency 13.8000 18.0000 5.0000 22.0000 4.0000 20.0000
Purchase 3200.0000 2000.0000 5000.0000 1500.0000 6000.0000 1800.0000
=== Clustered Instances ===
Inst# UsageTime LoginFrequency Purchase Cluster
1 120.000 18.000 2000.000 0
2 60.000 5.000 5000.000 1
3 140.000 22.000 1500.000 2
4 50.000 4.000 6000.000 3
5 130.000 20.000 1800.000 4
Status



Q4. Agricultural Crop Monitoring

Answer

The agricultural dataset contains three attributes:

- Farm rainfall
- Soil fertility score
- Crop yield

The attributes have different scales. Rainfall ranges approximately from **280–520 mm**, soil fertility ranges from **3–9**, and crop yield ranges from **65–90 tons**.

Before Normalization

Without normalization, the rainfall values have a much larger numerical range than soil fertility. Distance-based clustering algorithms may therefore give excessive importance to rainfall.

As a result, farms may be grouped mainly according to their rainfall values rather than considering all three agricultural characteristics equally.

For example:

- F1 and F4 have relatively low rainfall and high soil fertility.
- F2 and F5 have high rainfall and low soil fertility.
- F3 has intermediate values.

After Normalization

Normalization converts the attributes into comparable scales, commonly between **0 and 1**. After normalization, rainfall, soil fertility, and crop yield contribute more equally to the distance calculation. Therefore, the clusters represent overall agricultural similarity instead of being dominated by rainfall.

Possible groups include:

- **High fertility and high yield farms:** F1 and F4
- **High rainfall and lower fertility/yield farms:** F2 and F5
- **Intermediate agricultural conditions:** F3

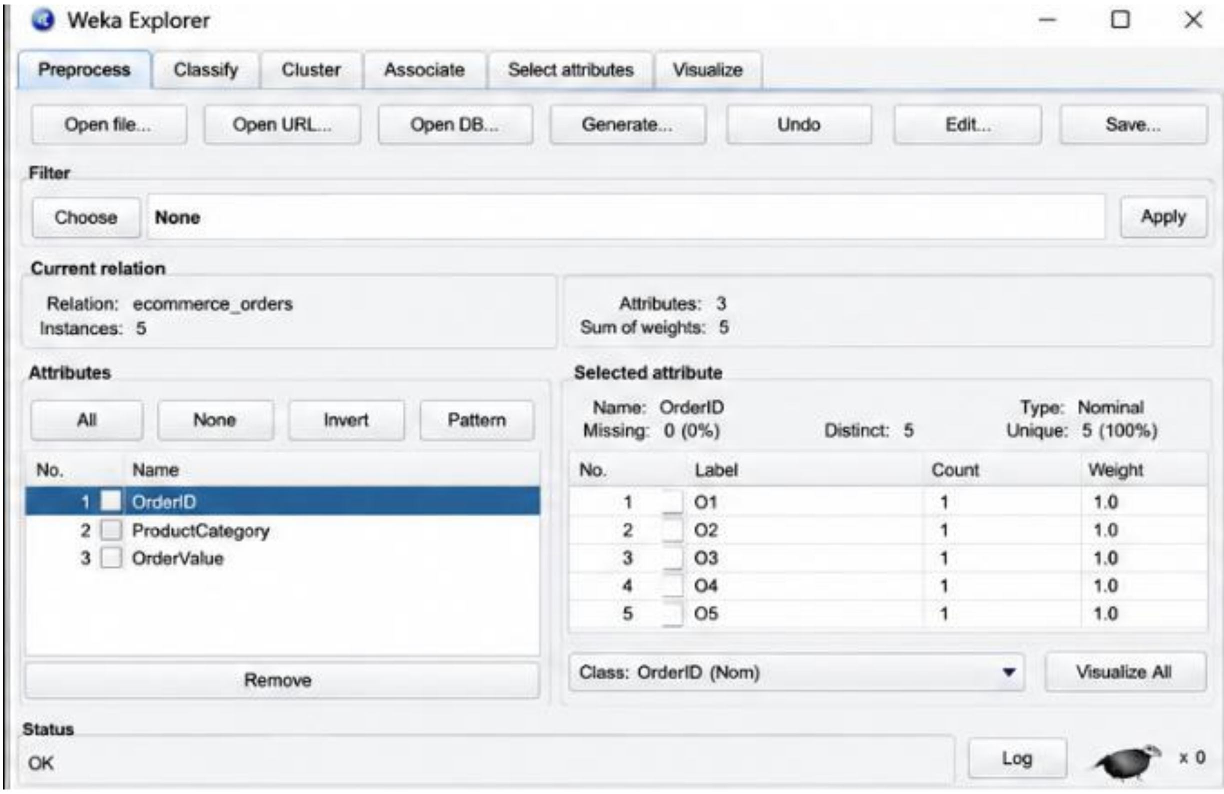
Importance of Normalization

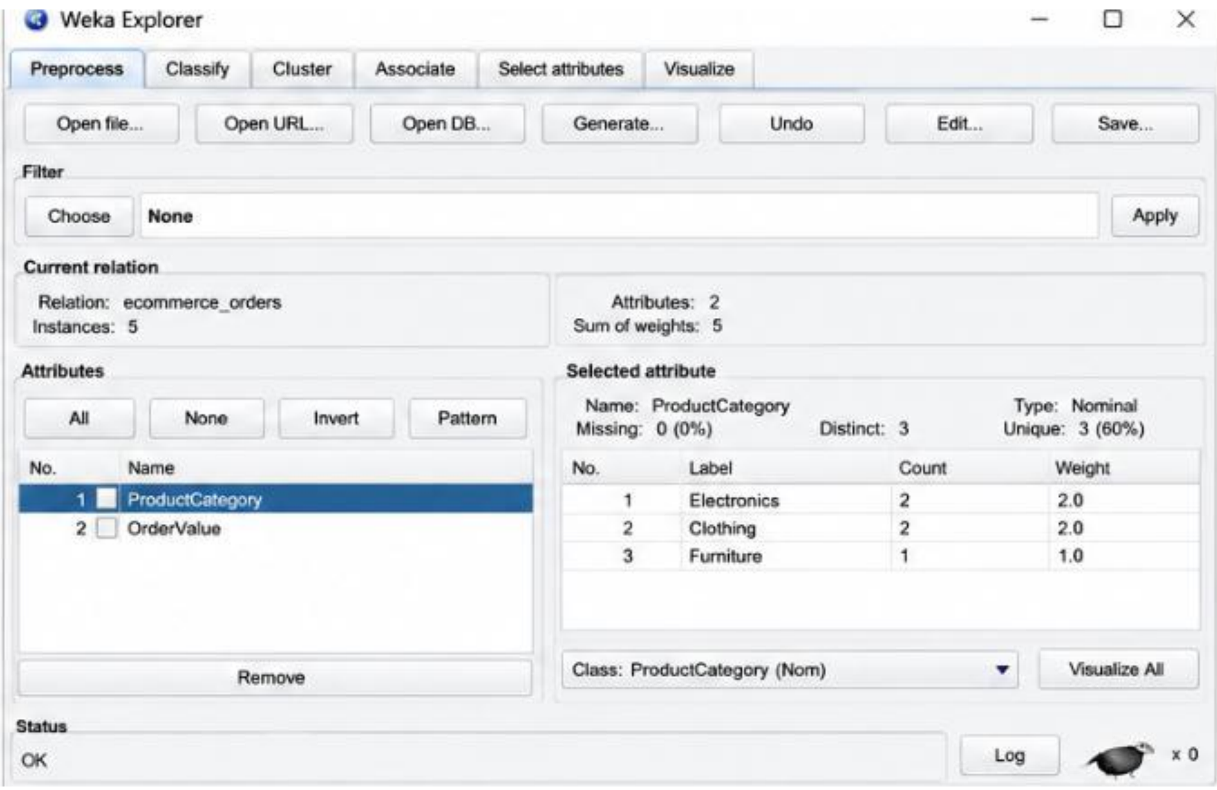
Normalization:

1. Prevents large-scale attributes from dominating clustering.
2. Gives each feature a fair contribution.
3. Produces more meaningful clusters.
4. Improves the reliability of distance-based algorithms.
5. Helps researchers make better agricultural decisions.

Conclusion

Normalization is important because it ensures that all agricultural attributes contribute fairly to cluster formation. Therefore, the clusters obtained after normalization are generally more reliable and meaningful.

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Attributes: 2
Sum of weights: 5
Attributes
All
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Invert
Pattern
No. Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
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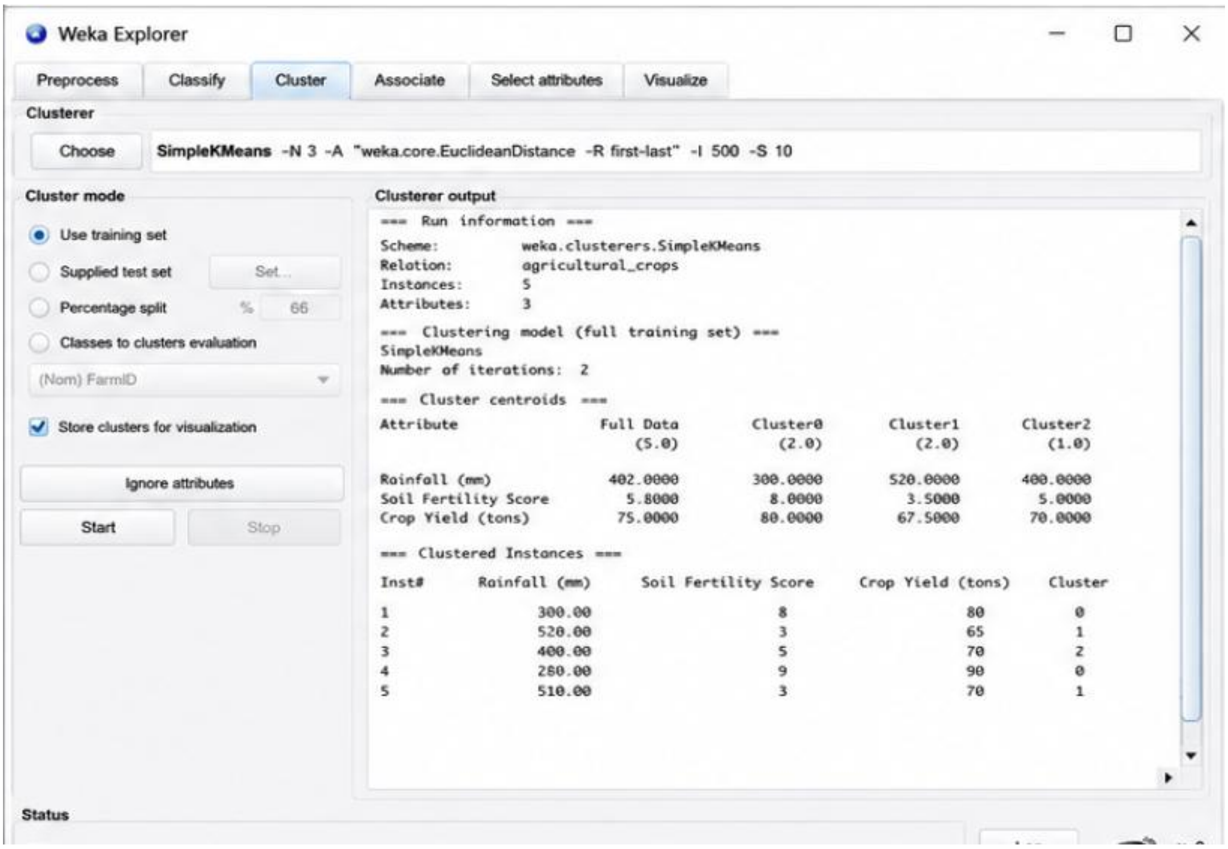
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Classify
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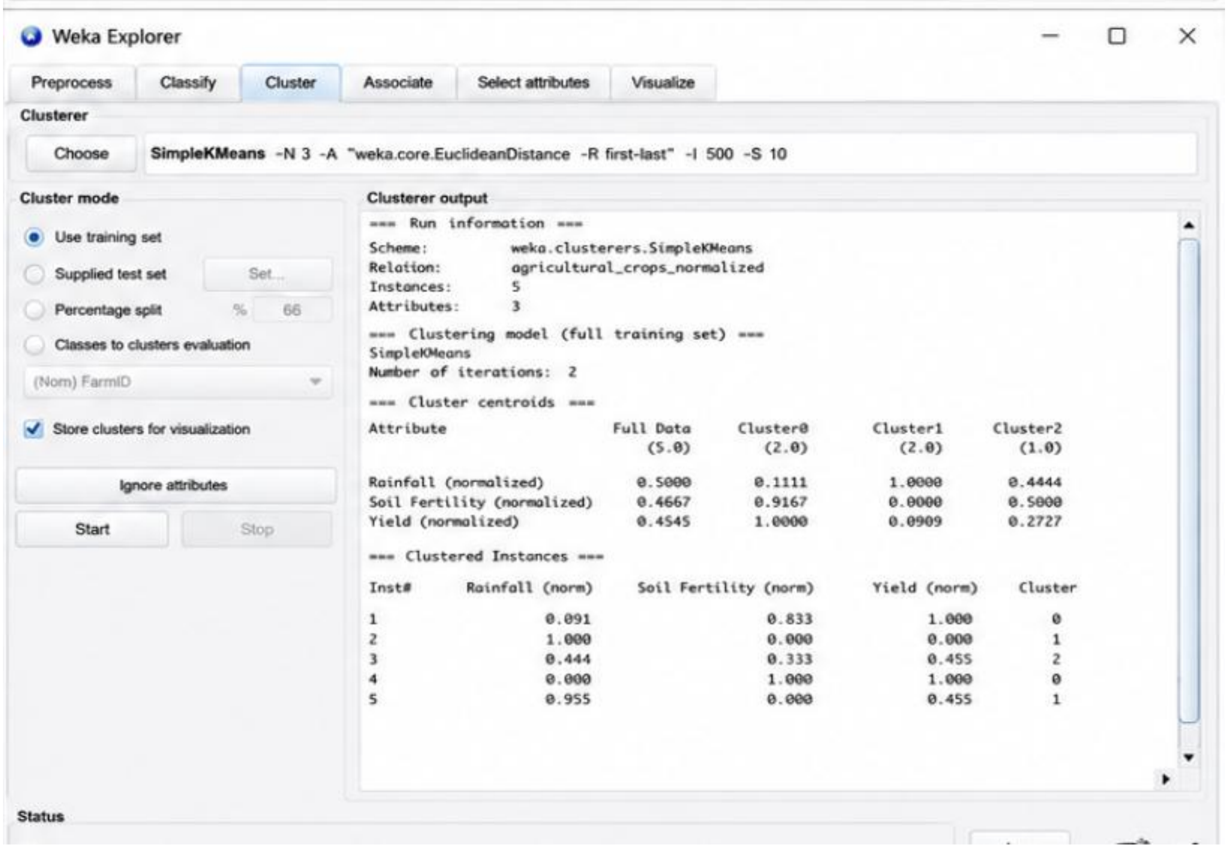
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) FarmID
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: agricultural_crops
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2
(5.0) (2.0) (2.0) (1.0)
Rainfall (mm) 402.0000 300.0000 520.0000 400.0000
Soil Fertility Score 5.8000 8.0000 3.5000 5.0000
Crop Yield (tons) 75.0000 80.0000 67.5000 70.0000
=== Clustered Instances ===
Inst# Rainfall (mm) Soil Fertility Score Crop Yield (tons) Cluster
1 300.00 8 80 0
2 520.00 3 65 1
3 400.00 5 70 2
4 280.00 9 90 0
5 510.00 3 70 1
Status



Q5. Employee Performance Evaluation

Answer

The given silhouette scores are:

Number of Clusters	Silhouette Score
K = 2	0.48
K = 3	**0.68**
K = 4	0.52

A silhouette score measures how well data points fit within their own cluster compared with other clusters. A **higher silhouette score indicates better-defined and more separated clusters**.

Among the three models:

- K=2 has a score of **0.48**.
- K=3 has the highest score of **0.68**.
- K=4 has a score of **0.52**.

Therefore, **K = 3 is the most suitable number of clusters**.

Employee Segmentation

The three clusters can represent different performance groups:

- **Cluster 1 – High performers:** High productivity, attendance, and skill scores.
- **Cluster 2 – Average performers:** Moderate productivity, attendance, and skill scores.
- **Cluster 3 – Lower performers:** Lower productivity, attendance, and skill scores.

For example, E1 and E5 have high scores and are likely to belong to the high-performance group. E4 has relatively low scores and may belong to the lower-performance group. E2 and E3 may represent the middle-performance group.

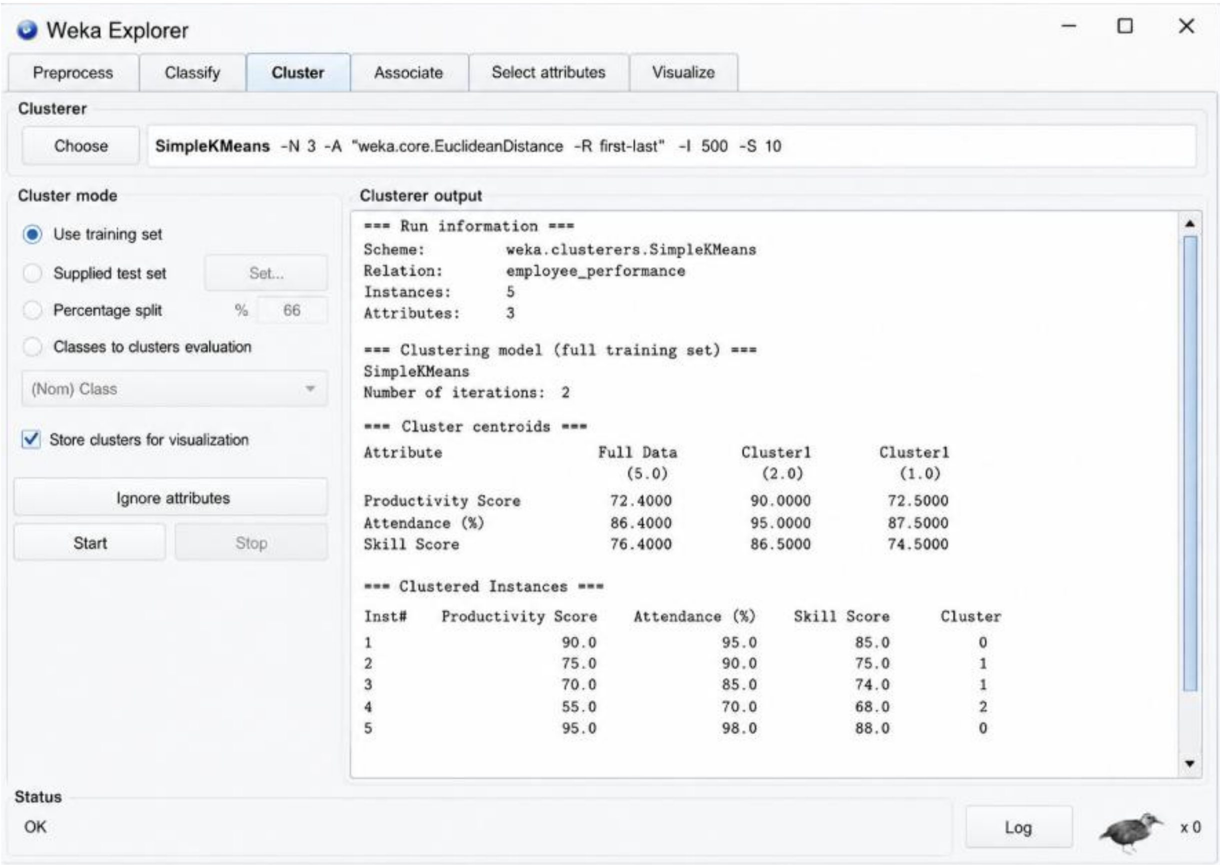
Importance for Employee Evaluation

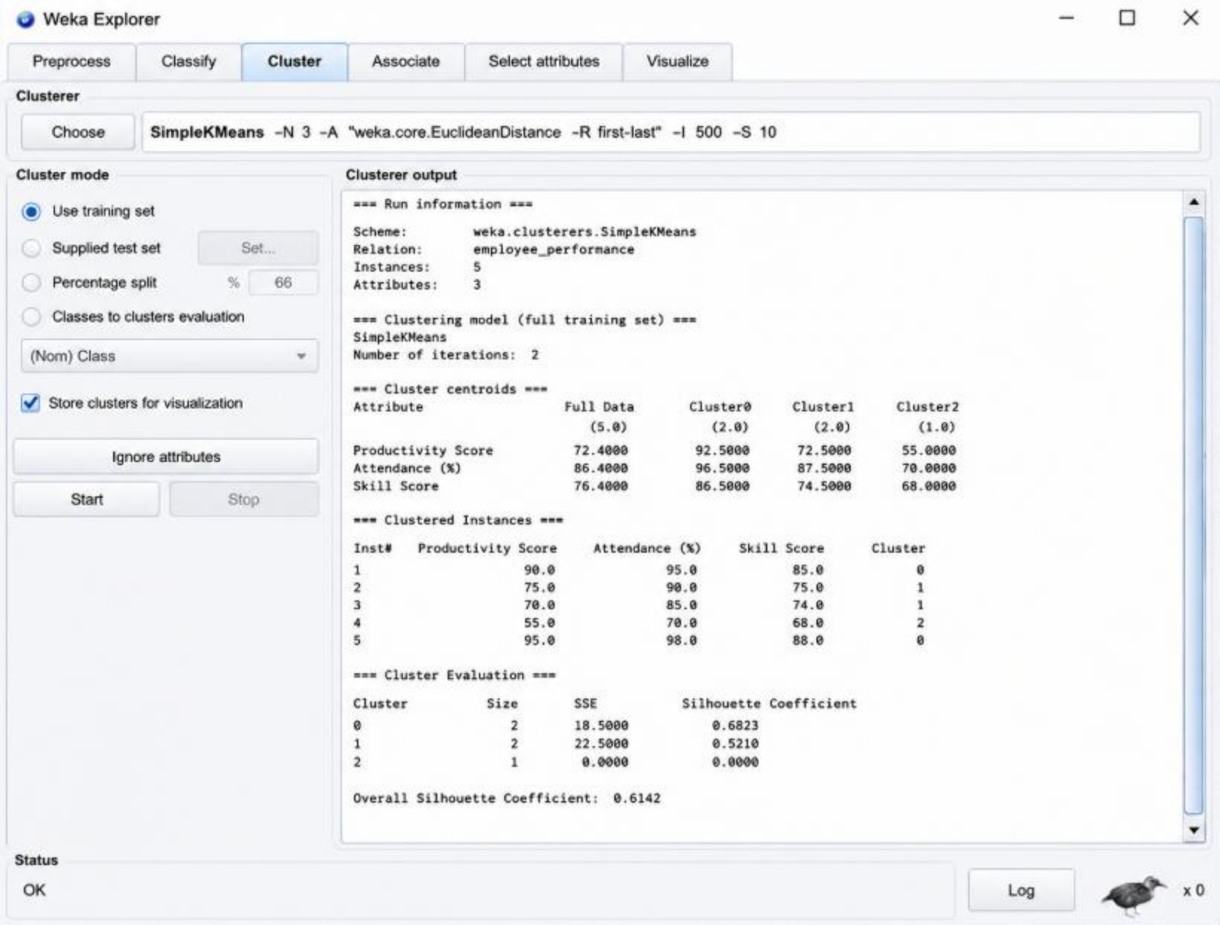
Using three clusters helps the company:

1. Identify high-performing employees.
2. Identify employees who may need training.
3. Understand different performance patterns.
4. Plan rewards and recognition.
5. Design targeted employee development programs.

Conclusion

The best model is **K = 3 because it has the highest silhouette score of 0.68**. It provides better-defined employee groups than K=2 or K=4 and can support effective performance evaluation and workforce management.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) Class
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: employee_performance
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2
(5.0) (2.0) (2.0) (1.0)
Productivity Score 72.4000 92.5000 72.5000 55.0000
Attendance (%) 86.4000 96.5000 87.5000 70.0000
Skill Score 76.4000 86.5000 74.5000 68.0000
=== Clustered Instances ===
Inst# Productivity Score Attendance (%) Skill Score Cluster
1 90.0 95.0 85.0 0
2 75.0 90.0 75.0 1
3 70.0 85.0 74.0 1
4 55.0 70.0 68.0 2
5 95.0 98.0 88.0 0
=== Cluster Evaluation ===
Cluster Size SSE Silhouette Coefficient
0 2 18.5000 0.6823
1 2 22.5000 0.5210
2 1 0.0000 0.0000
Overall Silhouette Coefficient: 0.6142
Status
OK
Log
x 0